

DNA REPLICATION

INTRODUCTION:

The important role of DNA in transfer of information in living cells, formulated by Francis Crick, is called the central dogma of molecular biology (Figure 20.1), which defines three major steps in the processing of genetic information.

- 1) The first is replication, the copying of parent DNA to form daughter DNA molecules having nucleotide sequences identical to those of the parent DNA.
- 2) The second step is transcription, the process in which the genetic messages in DNA are rewritten in the form of ribonucleic acid (RNA).
- 3) The third step is translation in which the genetic message coded by RNA is translated by the ribosomes into the protein structure.

This flow of genetic information from DNA to RNA to protein is called “**the central dogma**” of molecular biology.

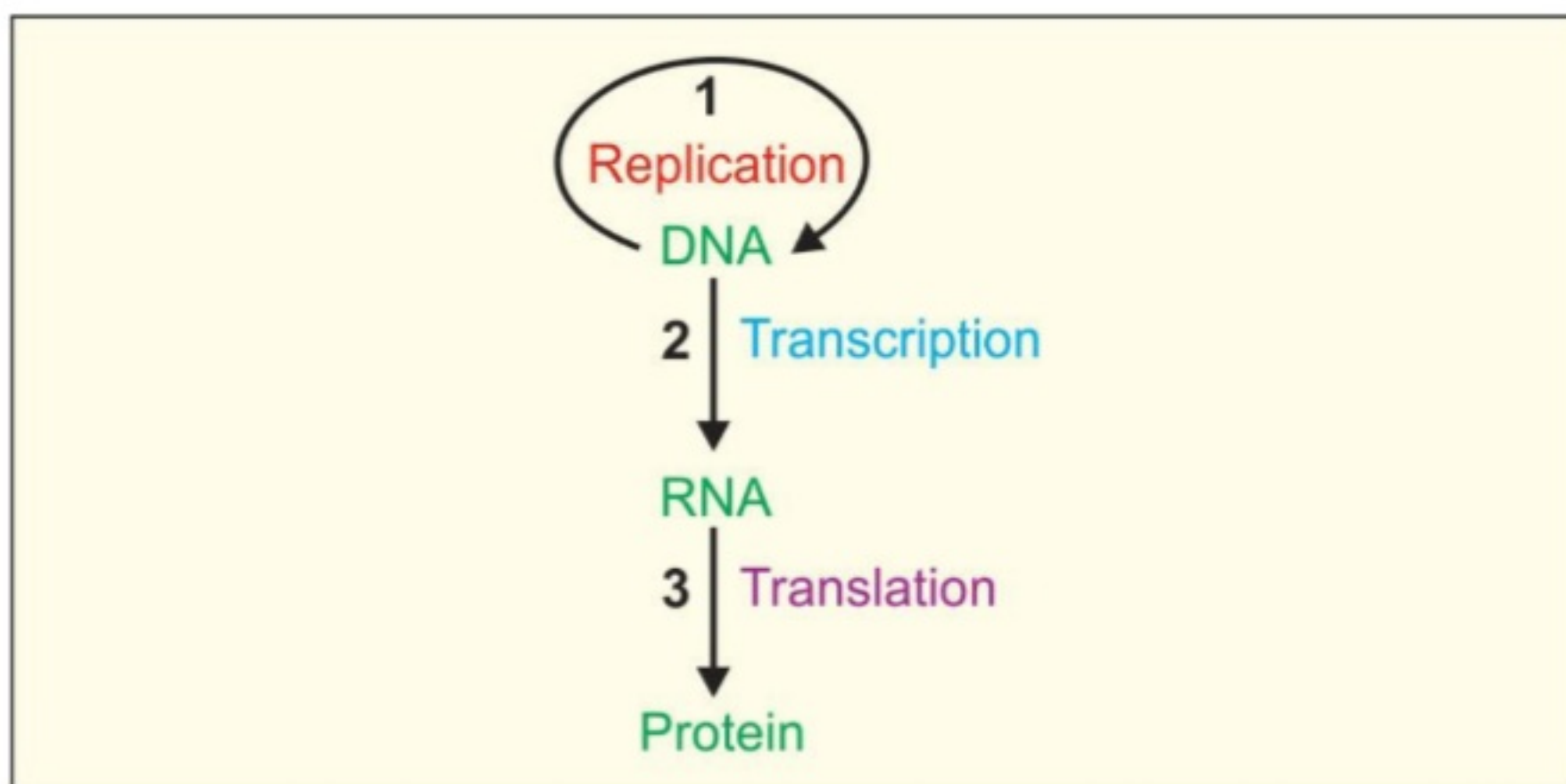


Figure 20.1: The central dogma of molecular biology

DNA REPLICATION (DNA SYNTHESIS):

DNA is a major store of genetic information. To transfer this genetic information from a parent cell to a daughter cell during cellular reproduction, the DNA must be duplicated. **The duplication or synthesis of DNA is called replication.**

This method of DNA replication results in semi conservative mechanism, in which each replicated duplex daughter DNA molecule contains one parent strand and one newly synthesized strand (Figure 20.2).

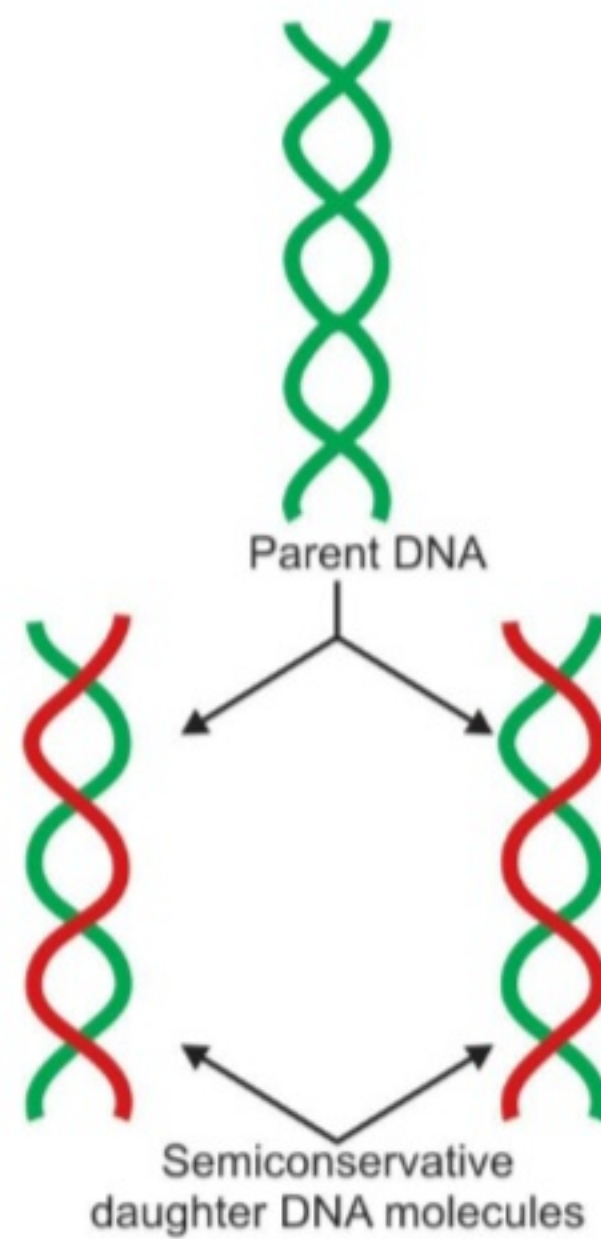


Figure 20.2: Semiconservative DNA molecules

PROKARYOTIC REPLICATION:

Replication in prokaryotes is much better understood than replication in eukaryotes. The basic requirements and components of replication are the same for prokaryotes as for eukaryotes. Therefore, an understanding of how prokaryotes replicate provides the understanding of how eukaryotes replicate.

- The DNA synthesis is catalyzed by enzyme called **DNA dependent DNA polymerase**. They are called DNA dependent as they require DNA template. They are more commonly called **DNA polymerases**. These polymerases, which are required for:
 - DNA chain elongation
 - DNA repair (5' - 3' exonuclease activity)
 - Proofreading (3' - 5' exonuclease activity).
- There are three types of polymerases in prokaryotes
 1. **DNA polymerase I (Pol I):** DNA polymerase I completes chain synthesis between Okazaki fragments on the lagging strand.
 2. **DNA polymerase II (Pol II):** Pol II is mostly concerned with proofreading and DNA repair.
 3. **DNA polymerase III (Pol III):** Pol III catalyzes leading and lagging strand synthesis.

- At least five DNA-polymerases exist in eukaryotic cells, α , β , γ , δ and ϵ (Table 20.1).

Table 20.1: A Comparison of prokaryotic and eukaryotic DNA polymerase		
Prokaryotic polymerase	Eukaryotic polymerase	Function
I	α	Gap filling and synthesis between Okazaki fragments of lagging strand
II	ϵ	DNA proofreading and DNA repair
	β	DNA repair
	γ	Mitochondrial DNA synthesis
III	δ	Leading and lagging strand synthesis

STAGES OF REPLICATION:

The process of replication can be divided into three stages.

- 1) Initiation
- 2) Elongation
- 3) Termination.

1) INITIATION:

- Initiation of DNA replication involves unwinding (separation) of two complementary DNA strands and formation of replicating fork.
- Unwinding occurs at a single, specific site at a particular DNA sequence on circular DNA of prokaryotes. The site is called **the origin of replication, "Ori"** where active synthesis occurs. This region is called **replicating fork** (Figure 20.3).
- Replication of double stranded DNA is bidirectional.
- In eukaryotes replication begins at multiple sites composed almost exclusively of A-T base pairs along the DNA helix and is referred to as a **consensus sequence**.

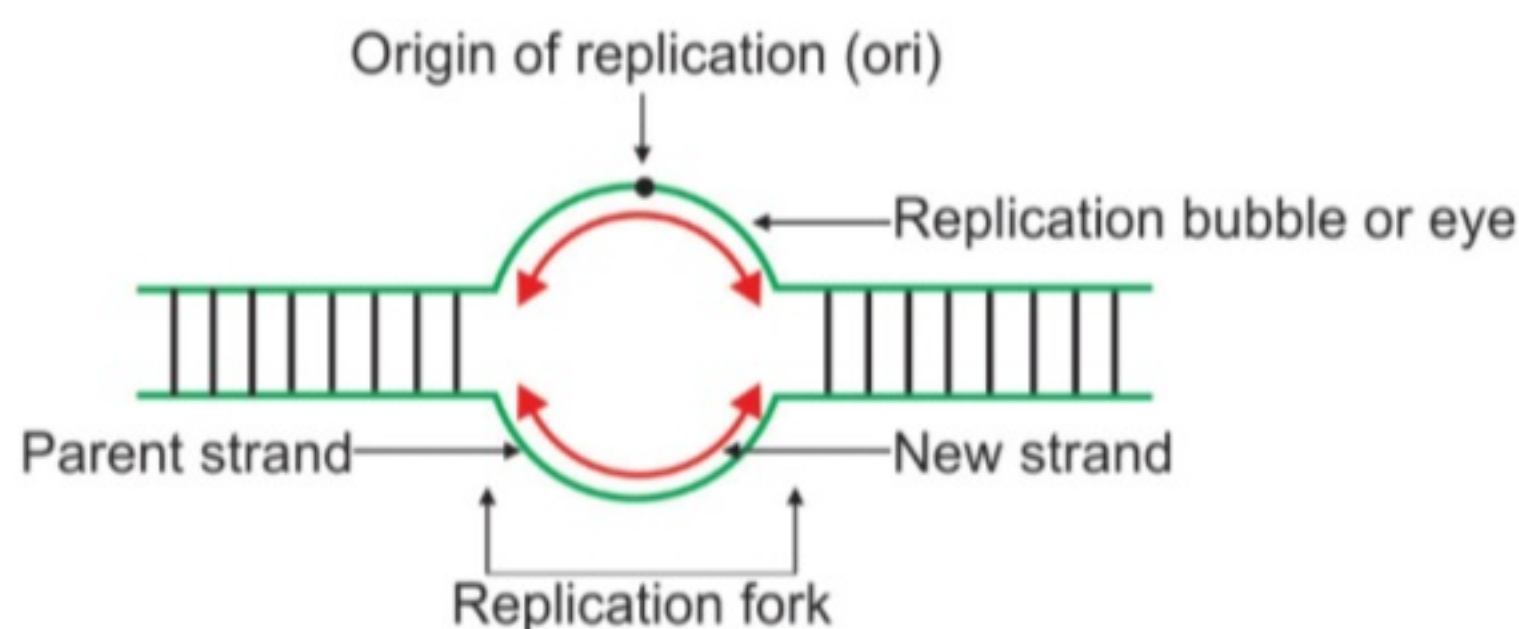


Figure 20.3: Formation of replicating fork

Steps involved in initiation:

1. First **DNA A protein** recognizes and binds to the “**ori**” of the DNA and successively denatures the DNA.
2. **DNA B protein (helicase)** then binds to this region and unwinds the parental DNA, and form a “V” where active synthesis occurs. This region is called the **replicating fork (Figure 20.4)**.
3. The stress produced due to unwinding by helicase is released by **topoisomerases** by cutting either one or both DNA strands.
4. The **Single stranded binding (SSB)** protein stabilizes the separated strands and prevents their reassociation.
5. To initiate the DNA synthesis by DNA polymerase III, it requires **RNA primer**. The **RNA primers** are short pieces of RNA (some 5-50 nucleotides in length) formed by the enzyme **primase** (RNA polymerase) using DNA as a template.

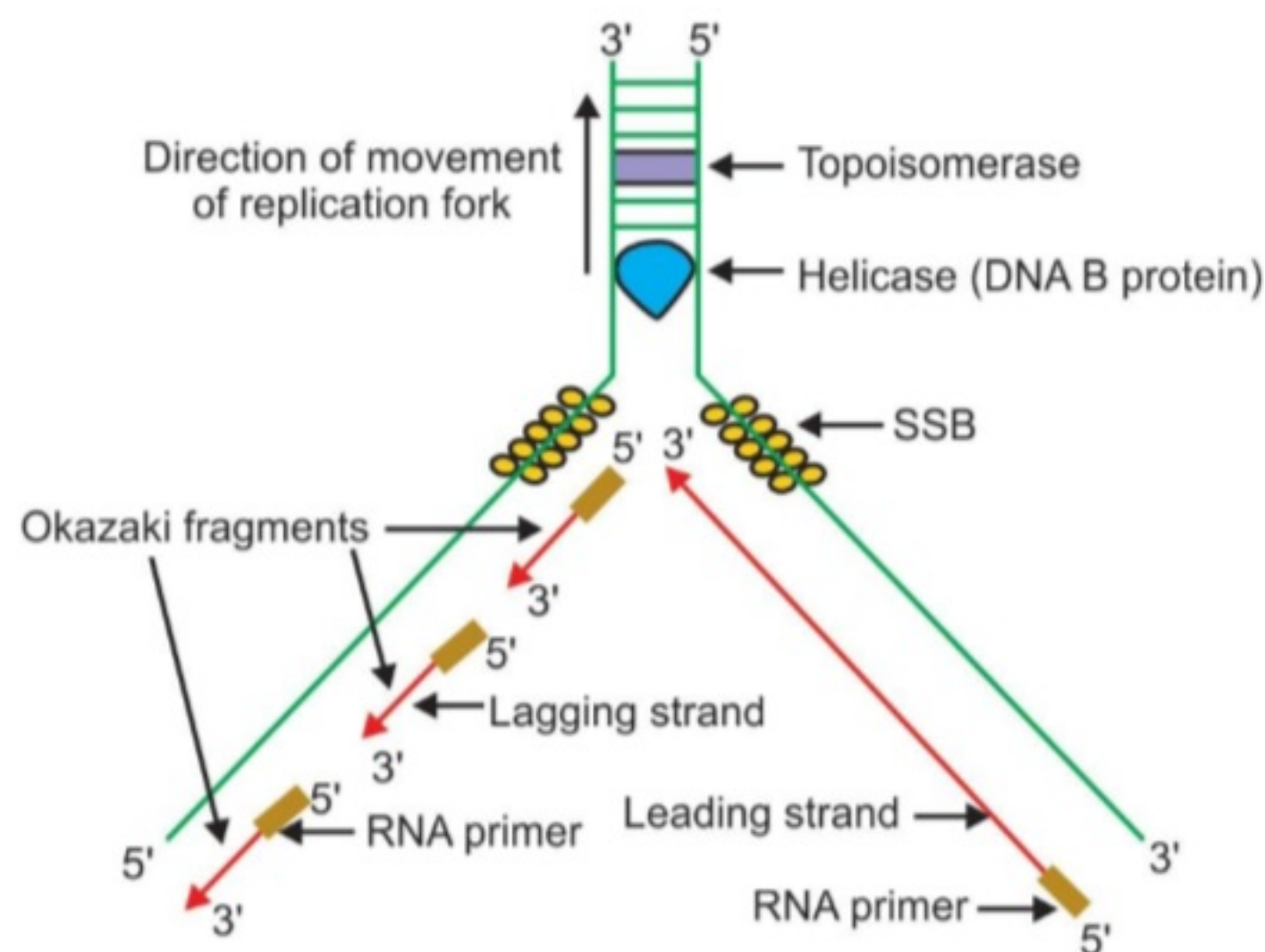


Figure 20.4: Replicating fork

2) ELONGATION:

- Once RNA primer has been synthesized at each of the replicating forks, a **DNA polymerase III** initiates the synthesis of new DNA strand by adding deoxyribonucleotide to the 3' end of the RNA primer. Thus, DNA polymerase III can synthesize a new chain only in the 5' to 3' direction. Both the DNA strands are synthesized simultaneously but in opposite direction one is in direction

towards the replication fork, the other in a direction away from the replication fork.

- The DNA chain which runs in the 3' → 5' direction is copied by polymerase III as a continuous strand, requiring one primer. This new strand is known as the **leading strand**.
- The DNA chain which runs in the 5' → 3' direction is copied by polymerase III as a discontinuous manner because synthesis can only proceed in the 5' to 3' direction. This new strand is known as the **lagging strand**. This requires numerous RNA primers. As the replication fork moves, RNA primers are synthesized at specified intervals. These RNA primers are extended by DNA polymerase III into short pieces of DNA called **Okazaki fragments**.
- Upon completion of lagging strand synthesis, the RNA primers are removed from fragments by **DNA polymerase I**. DNA Polymerase I also fills the gaps that are produced by removal of the primer leaving only a nick. It cannot join two polynucleotide chains together, an additional enzyme **DNA ligase** is required to perform this function. This enzyme catalyzes the formation of a phosphodiester bond to seal the Okazaki fragments.

3) TERMINATION:

Termination sequences, e.g. "**ter**", direct termination of replication. A specific protein, **ter binding protein**, binds these sequences and prevents the helicase (DNA B protein) from further unwinding of DNA and facilitates the termination of replication.

PROOFREADING:

- DNA is copied by DNA polymerase with high fidelity (accuracy). Incorrect nucleotides are incorporated with a frequency of one in 10^8 - 10^{12} bases, which could lead to **mutation**. But the error ratio during replication is kept at a very low level by specific process. This process is known as **proofreading**.
- Mismatches, occur more frequently but do not lead to stable incorporations because the all three **DNA polymerases** have **3' to 5' exonuclease** activity (proofreading activity).
- DNA polymerase I and II are known to excise mismatched nucleotides before the introduction of the next nucleotide.

EUKARYOTIC REPLICATION:

DNA replication in eukaryotic organisms resembles that in prokaryotic cells and proceeds by a mechanism similar to that of prokaryotic replication but is not identical. Some differences are given in Table 20.2.

Table 20.2: Some differences between prokaryotic and eukaryotic DNA replication

Features	Prokaryotes	Eukaryotes
RNA primer length	~50 nucleotides	9 nucleotides
DNA polymerase	Three types and designated by Roman numerals; I, II, III	Five types and designated by Greek numerals α , β , γ , δ and ϵ
Number of origins	Single	Multiple
Nucleotide length of Okazaki fragments of lagging strand	1000–2000 nucleotides	~200 nucleotides
Rate of replication	~500 nucleotides per sec	50 nucleotides per sec (10 times slower than prokaryotes)



THANK YOU

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